Herman Singh Umrao

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Professional Summary

Robotics and Edge AI engineer with hands-on experience building and deploying autonomous drones and mobile robots on real hardware. Currently working at PES University's Mobile and Autonomous Robotic Systems (MARS) Lab, developing ROS 2-based SLAM, perception, and navigation systems using IMU–LiDAR fusion and monocular vision. Strong in C++ and Python with deep experience across embedded platforms, Linux, and edge AI pipelines. Known for translating algorithms into reliable, resource-efficient robotic systems through rapid prototyping, testing, and iteration.

Key Technical Skills

- **Programming:** C++ (primary), Python (rapid prototyping)
- **Edge AI & ML:** TensorFlow Lite, ONNX, YOLO, PyTorch, TensorFlow
- **Robotics & Autonomous Systems:** ROS 2, SLAM, Isaac ROS, Gazebo, RViz2
- **Workflow:** Real-time debugging, sensor calibration, hardware bring-up
- **Computer Vision:** OpenCV, MediaPipe, Dlib, Monocular Depth Estimation
- **Embedded Systems:** STM32, Arduino, ESP32, Pixhawk, Raspberry Pi
- **Hardware Design:** FreeCAD, Blender
- **Systems:** Linux (Arch & Ubuntu), Docker, ROS 2 Humble & Jazzy
- **Manufacturing:** FDM 3D printing, rapid prototyping

Experience:

Research Intern |
Mobile & Autonomous Robotic Systems Labs @ PESU — Nov 2024 - Present

- Built and deployed autonomous drones and mobile robots using ROS 2
- Implemented SLAM, visual odometry, and navigation pipelines for GPS-denied environments
- Integrated IMU–LiDAR fusion and monocular vision on real robotic platforms
- Tested, debugged, and iterated perception and navigation stacks on hardware
- Designed drone testing jigs and automated PID tuning workflows
- Worked closely with cross-functional teams and faculty to deliver working systems

ROS for Autonomous Systems, PES University — Aug 2025 - Dec 2025

- Assisted in developing course slides, laboratory exercises, and instructional media for ROS-based robotics
- Supported students during labs and project work under faculty supervision
- Contributed academic materials published on the official university website

Founder & Club Head – LinuxLegion Jan 2024 – Jun 2025

- Co-founded and led a community of 50+ students focused on Linux and open-source technologies
- Organized hands-on workshops, technical talks, and peer-learning sessions
- Promoted adoption of open-source tools and collaborative engineering practices

Software Development Intern | Tech Mahindra Cerium Pvt Ltd Jun 2024 - Jul 2024

- Developed and demonstrated an AI-based human emotion detection system
- Applied computer vision and machine learning techniques for real-time inference
- Presented working prototypes to internal technical teams

Research Intern | Jan 2024 - May 2024
Center for Innovation and Entrepreneurship @ PESU in collaboration with Intel

- Worked with Intel Simics full-system simulator for chipset and platform-level development
- Gained exposure to low-level system architecture, including Intel CISC and RISC-V platforms
- Assisted in simulation-based experimentation and technical analysis

Teaching Assistant | Aug 2023 - Dec 2023
SOC Design and Computer Architecture Dept, PES University

- Created audiovisual learning content to support undergraduate coursework
- Materials were published on the official university platform
- Assisted students under faculty guidance (Dr. Jayashree, Nagegowda K. S.)

Selected Engineering Projects

- Built monocular visual odometry pipelines and explored 3D scene reconstruction
- Developed a full autonomous navigation stack for ROS 2 robots
- Implemented SLAM-based navigation using LiDAR and stereo cameras
- Designed and built drone testing jigs and automated PID tuning systems
- Competed in robotics competitions including RoboWars and Robosoccer

Previous projects:

- Edge AI CCTV Cameras running minimalistic AI powered CCTV systems based on Pi Zero hardware.
- Programming arm robots(NED2 Niryo), humanoid(NAO), drone programming
- 3D mapping using IMU-LiDAR fusion
- 2d LiDAR based AMR
- Qt_turtlebot localizer, A GUI interface for turtle-bot re-localization
- Drone motor thrust measuring JIG
- Modular drone based delivery system
- RF-based file finder
- Docker-Hadoop, Containerization and configuration of all Big-Data tools

Education

- Bachelor of Technology - Computer Science (AI & ML)

PES University | Sep 2022 - Sep 2026

- Senior Secondary and High School - Physics, CS, Mathematics, Chemistry, English

HAL Public School | june 2008 – mar 2022